

# Case Study: NLTTA – Translating Romanized Nepali to English for News Retrieval

## Overview

In Nepal, many users communicate digitally using Romanized Nepali, a transliterated form of the language often mixed with English. Traditional search engines struggle with such informal, code-mixed inputs. To address this, we developed NLTTA (Nepali Language Translation and Transliteration Automation) — a full-stack system that translates Romanized Nepali queries into English and retrieves relevant news articles using neural machine translation and probabilistic retrieval.

## Key Features

- Translation Model: Fine-tuned mBART model trained on a parallel corpus of Romanized Nepali and English.
- Retrieval Algorithm: BM25 algorithm applied on a pre-indexed corpus of English news articles.
- Deployment: REST API using Django to serve real-time translations and search results.
- Frontend: Built using Next.js with a clean and responsive interface for entering queries and viewing results.

## How It Works

1. User Input: User enters a search query in Romanized Nepali.
2. Translation: The NLTTA model translates the query into English.
3. Search: The translated query is ranked using BM25 against a news database.
4. Results: Top-matching articles (title, image, summary) are displayed.

## Technical Stack

- Languages: Python, JavaScript
- Libraries: PyTorch, Hugging Face Transformers
- Frameworks: Django (backend), Next.js (frontend)
- IR Engine: BM25 (with custom TF/IDF optimization)
- Database: MySQL
- Deployment Tools: REST API (Postman, Docker-ready)

## Evaluation

- BLEU Score: 39.84 — comparable to larger models but with significantly less compute.
- BERT Score: 0.9613 — demonstrating strong semantic understanding.
- Latency: Sub-3 second response time for most queries on limited hardware.

- Coverage: Supports Devanagari, Romanized, and mixed Nepali-English inputs.



***BLEU and BERT score along with model Parameter size***

## Impact

- Inclusivity: Empowers partially literate and non-English-speaking users in rural Nepal to access English news.
- Scalability: Can be extended to e-commerce, government, or educational platforms.
- Open-Source Friendly: Uses free datasets and cloud-based training (Google Colab), making it accessible and low-cost.